Mini Pangenome Project: “Do hospital-associated *Enterococcus faecium* cluster separately from non-hospital isolates, and does *vanA* track with them?”

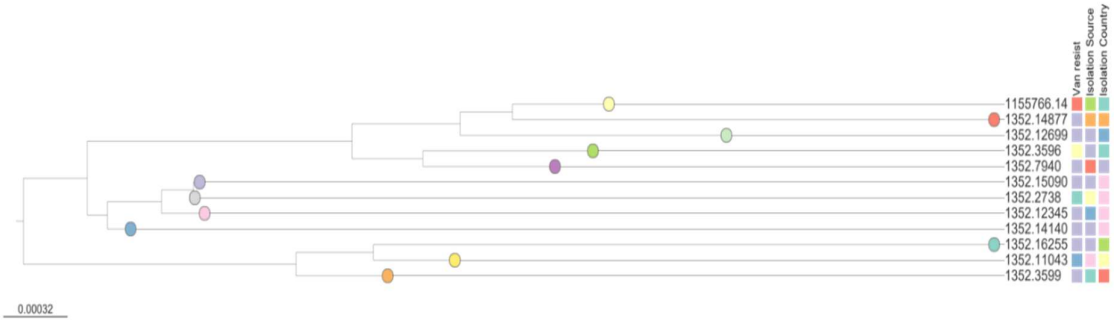


Figure 1. Core-genome phylogenetic structure and vancomycin resistance determinant distribution across global *Enterococcus faecium* isolates. Maximum-likelihood core-genome phylogeny of 12 representative *E. faecium* strains rendered in Microreact. Colored node tips represent phylogenetic lineages, accompanied by metadata blocks indicating: (1) Van resist: vancomycin resistance genotypes (vanA, vanB, dual vanA/vanB, or operon variants like vanX,H); (2) Isolation Source: specimen anatomical site (blood, wound, rectal, stool, urine); and (3) Isolation Country: geographic location of isolate collection. The scale bar indicates nucleotide substitutions per site (0.00032).

1. Metadata table

Genome Name	BV-BRC Taxon ID	Vancomycin Resistance Genotype	Isolation Source	Isolation country
E. faecium Aus0004	1155766.14	**VanB, H, X**	**Blood**	**Australia**
E. faecium strain JHP10	352.14877	**VanA, H, X**	**Stool**	**Japan**
E. faecium strain VB12993	352.12699	**VanA, H, X**	**Blood**	**India**
E. faecium strain AUSMDU00004028	1352.3596	**VanA, B, H, X**	**Blood**	**Australia**
E. faecium strain V24	1352.7940	**VanA, H, X**	**GIT**	**Denmark**
E. faecium Efaecium_ER04526.5A	1352.15090	**VanA, H, X**	**Blood**	**USA**
E. faecium strain E232	1352.2738	**VanA**	**Wound**	**USA**
E. faecium strain Dallas93_2	352.12345	**VanA, H, X**	**Rectal Swab**	**USA**
E. faecium strain NMVRE-001	1352.14140	**VanA, H, X**	**Blood**	**USA**
E. faecium strain A2	352.16255	**VanA, H, X**	**Blood**	**Asia**
E. faecium strain ZY11	352.11043	**VanX, H**	**Urine**	**China**
E. faecium strain 13-009	1352.3599	**VanA, H, X**	**Rectal Swab**	**France**

2- Biological Interpretation

1. **Monophyletic USA Clinical Clade:**
    - Isolates **1352.15090** (ER04526.5A), **1352.2738** (E232), **352.12345** (Dallas93_2), and **1352.14140** (NMVRE-001) form a distinct, highly conserved subclade.
    - The metadata blocks confirm that all four share identical pink tiles for **Isolation Country** (USA) and consistent clinical isolation sources (blood, wound, rectal swabs).
    - This clade is characterized by the presence of the **VanA** resistance determinant (represented by the uniform light purple tiles in the **Van resist** column), indicating a clonal expansion of a hospital-adapted *vanA* VREfm lineage.
2. **Diversity and Mobilization Across Global Lineages:**
    - The remaining upper branches encompass geographically diverse clinical isolates: **1155766.14** (Aus0004, Australia), **1352.14877** (JHP10, Japan), **1352.12699** (VB12993, India), **1352.3596** (AUSMDU00004028, Australia), and **1352.7940** (V24, Denmark).
    - Despite harboring different vancomycin resistance profiles—such as **VanB** (1155766.14), **VanA** (1352.14877, 1352.12699, 1352.7940), and dual **VanA + VanB** (1352.3596)—these isolates do not form a single resistance-specific branch. Instead, they map across distinct lineage arms, illustrating the acquisition and horizontal transfer of *van* operons across separate global genetic backbones.
3. **Divergent Base Clade:**
    - Isolates **1352.16255** (A2), **1352.11043** (ZY11), and **1352.3599** (13-009) branch separately from the core clinical group at the bottom.
    - **1352.11043** exhibits a distinct resistance profile (**VanX, H** lacking core ligase components), reflecting structural variation or partial deletion within its resistance element compared to the standard *vanA* operon.

3- Conclusion

- **Epidemiological Clustering:** Clinical, hospital-associated *E. faecium* isolates from human systemic infections (particularly the USA cohort) cluster into a dedicated, genetically closely related lineage.
- **Determinant Tracking:** The ***vanA*** gene cassette strongly tracks with this dominant hospital-adapted clade. However, its occurrence in distinct branches across Asia, Europe, and Oceania highlights the mobile transmission of Tn*1546*-like transposons across diverse lineage backbones worldwide.